ENGINEERING STUDIES

TEST 2 SCOPE 2026

SUBJECT	LEVEL	MODULE
Engineering Drawing	N2	Module 6:First Angle Orthographic projection (Sectional Drawing); Module 7: Isometric Drawing, Module 8: Third Angle Orthographic Projection, Module 9: Curves and Interpenetration
Engineering Science	N2	MODULE 1:DYNAMICS MODULE 2: STATICS MODULE 3:ENERGY AND MOMENTUM MODULE 4: WORK, POWER AND EFFICIENCY (EXCLUDING FORCE DISTANCE GRAPHS AND WORK DONE ON AN INCLINED PLANE) MODULE 5: MECHANICAL DRIVES AND LIFTING MACHINES(MECHANICAL DRIVES ONLY) MODULE 6: FRICTION(DEFINITIONS AND CALCULATIONS ON COEFFICIENT OF FRICTION ONLY)
Mathematics	N2	MODULE 2: Mensuration; MODULE 3: Factorisation; MODULE 4: Exponents & Logarithms; MODULE 5:Equations & Manipulation of formulae; MODULE 6: Trigonometry; MODULE 7: Graphs
Instrument Trade	N2	NON ELECTRICAL TEMPERATURE MEASUREMENT LEVEL MEASUREMENT
Electrical Trade		AC THEORY: Power in AC circuits. Single and Three Phase Circuits. MODULE 2: Conductors,Insulators and Cables, Module 4: Switchgear and Protective Devices. MODULE 5: Batteries.
Motor Trade N2	N2	MODULE 2: GEAR BOX, MODULE 3: CLUTCH, MODULE 4: DRIVE SHAFT

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Bricklaying and Plastering	N2	Module 7:concrete masonry walls Module 8:Terrazzo floors Module 9: Wall and floor tiling Module 10:Fireplace and flues Module 11:Arches Module 12 Plain and fluted classical columns Module 13: Lathing Module 14: Ceiling plaster mouldings Module 15:Roof cladding materials Module 16: Pavings
Plumbing Trade	N2	1. Cold water supply 2. Hot water supply 3. Dranaige 4. Metal sheet (Drawing)
Fitting and Machining	N2	health and safety, couplings, bearing, fits and limits, v belts, lubrication, gear drives
SUBJECT	**LEVEL**	**MODULE**
Building and Civil Technology	101	Module 6: Masonry Module 7: Structural elements Module 8: Water supply and damage Module 9: Electrical work Module 10: Finishing Module 11: External works Module 12: Cost estimation
Building Drawing	101	Module 3: Floors and Foundation, Module 4: Roofs, Module 5: Stairs
Electrotechnology	101	MOULE 2: DC Machines. MODULE 3: DC Generators. MODULE 4: DC Motors. MODULE 5: Efficiency Testing of DC machines.
Engineering Drawing	101	Freehand Drawing 2 Point Perspective Sectional Drawing Detailed Drawing
Engineering Science	101	MODULE 1: MOTION, POWER AND ENERGY MODULE 2: MOMENTS(BEAMS) MODULE 3: FORCES(EXCLUDING SIMPLE FRAMEWORKS) MODULE 4: FRICTION(HORIZONTAL PLANE ONLY) MODULE 6: HYDRAULICS(HYDRAULIC PRESSES ONLY)
Industrial Electronics	101	1. Kirchhoff's laws 2. Diode applications 3. Operational amplifiers 4. AC - Theory
Mathematics	101	MODULE 1: Factors & Fractions; MODULE 2: Remainder & Factor Theorem; MODULE 3: Exponents, Surds & Logarithms; MODULE 4: Equations, Manipulation of Formulae & Word Problems; MODULE 5: Graphs; MODULE 6: Coordinate Geometry; MODULE 7: Limits & 1st Derivatives
Mechanotechnology	101	Module 4: Pumps Module 5 . Bearings Module 6: Internal combustion engines. Module 7 Cranes. Module 8 Brakes

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Chemistry	101	Module 1:Chemical bonding, Module 2:Chemical geometry, Module 3:Acids and bases, Module 4: Descriptive chemistry of hydrogen and carbon
Analytical Chemistry	101	Errors in chemical analysis, Random errors in chemical analysis, Statistical data treatment and evaluation.
Communication	101	1.Visual Literacy, 2. Structuring and formating conventions in texts, 3. Workplace texts: Agendas and Minutes, 4. Reading for different purposes,5. Using Reference skills, 6. Interpreting visual texts, 7. The writing process, 8. Correcting and editing, 9. Writing for the workplace (Report writing).
SUBJECT	**LEVEL**	**MODULE**
Building Administration	N4	Module 5: Building materials Module 6: Supports to structures Module 7: Scaffoldings Module 8: Lifts and cranes Module 9: Programming of work
Building and Structural Construction	N4	1. Foundation and Basements 2. Brickwork 3. Archs 4. Steel work 5. Some Theory
Chemical Plant Operations	N4	Module 3: Materials; Module 4: Industrial Chemical Gases Processes; Module 5: Organic Processes
Chemistry	N4	Module 1:Chemical bonding, Module 2:Chemical geometry, Module 3:Acids and bases, Module 4: Descriptive chemistry of hydrogen and carbon
Digits	N4	MODULE 2: NUMBER SYSTEM MODULE: 3 FURTHER CODES MODULE 4: LOGIC GATES MODULE 5: BOOLEAN ALGEBRA MODULE 9: ELEMENTARY FLIP-FLOPS MODULE 10: COUNTERS
Electrotechnics	N4	MODULE 2: DC Machines, DC Motors, DC Generators, MODULE 3: AC Circuits. RLC Circuits. MODULE 4: Transformers.
Engineering Science	N4	MODULE 1: KINEMATICS MODULE 2: ANGULAR MOTION MODULE 3: DYNAMICS MODULE 4 : STATICS MODULE 5: HYDRAULICS(HYDRAULIC PRESSES ONLY)
Industrial Electronics	N4	SUPERPOSITION,AC CIRCUITS,POWER SUPPLIES,ELECTRONIC POWER CONTROL,OP AMPS
Mathematics	N4	MODULE 4: TRIGONOMETRY ,MODULE 5 ;GRAPHS MODULE 6;LIIMITS AND DIFFERENTIATION

ENGINEERING STUDIES

TEST 2 SCOPE 2026

Mechanical Draughting N4	N4	Loci Cam Profiles Sectional Drawing Detailed Drawing
Mechanotechnics	N4	module .1 Workshop layout, Module 9 .Metal cutting machines, Module 5 Gear drives , Module 8 .Hydraulics systems
Quantity Surveying	N4	Module 4 : SI Units & Abbreviations from projects, Module 4.2 : Superstructure Project Measurements or 4.6 Plain Opening Project Measurements, Module 5 : Working Up Process & Theory, Module 6 : Price Analysis & Cost Estimates.
Instruments	N4	1. Non electrical temperature measurement 2. Electrical temperature measurement 3. Electrical Pressure measurement 4. Pyrometers 5. Telemetering
SUBJECT	**LEVEL**	**MODULE**
Building Administration	N5	Module 5: Supports fot structures Module 6: Mass haul diagrams Module 7: Variations ,omissions and handing over Module 8: Statutory bodies
Building and Structural Construction	N5	1. Beam design (Reactions , Shear force , Bending moment & Ze). 2. Centroid & Moment of Inertia 3. Bolted Joints 4. Frames 5. Bending scheduke 6. Drawings
Chemical Plant Operations	N5	MODULE 3: Separators, Cyclones and Mixers ; Module 4: Industrial Chemical Process
CHEMISTRY	N5	Module 1: INTRODUCTION TO ORGANIC CHEMISTRY(Isomerism and nomenclature) , Module 2: HYDROCARBONS(PROPERTIES, PREPARATIONS AND REACTIONS : ALKANES,ALKENES,ALKYNES,ALCOHOLS, ALDEHYDES AND KETONES). MUST KNOW COMMON NAMES
Digits	N5	1.MODULE 1: A/D AND D/A CONVERTERS 2.MODULE 4: K-MAPS 3. MODULE 5: CODE CONVERTERS MODULE 7,8,9: NUMBER SYSTEM MODULE10: PARTLY LINE CCT MODULE 11: COMPUTER ARCHITECTURE

ENGINEERING STUDIES

TEST 2 SCOPE 2026

Electrotechnics	N5	1 DC MACHINES 2 AC THEORY 3 TRANSFORMERS 4 MEASURING POWER THREE PHASES SYSTEMS 5 AC MACHINES
Engineering Physics	N5	module 2: Heat, module 3: Light, Module 4: Magnestism
Industrial Electronics	N5	1. Integrated circuits 2. Transistors 3. Operational amplifiers 4. Phase control 5. Transducers
Mathematics	N5	MODUEL 1 ;LIMITS ,MODULE 2;DIFFERENTIATION,MODULE 4;INTEGRATION
Mechanical D& Design	N5	cottered joint, lozenge joint, power transmission
Mechanotechnics	N5	Conveyour belt, Bucket Elevators, Endless rope haulage , Elevators
Power Machines	N5	1 THERMODYNAMICS 2 AIR COMPRESSORS 3 STEAM PLANT 4 STEAM CONDENCERS 5 PUMPS
Quantity Surveying	N5	Module 2 : INTERIM valuations & theory Module 3.4: R.C. Structure OR Module3.3: Garden walls, Module 4: Cost Estimates Calculations & Theory
Instruments	N5	FLOW , DENSITY, HUMIDITY, PH MEASUREMENT
SUBJECT	**LEVEL**	**MODULE**
Digital Electronics	N6	NUMBER SYSTEM , LOW LEVEL PROGRAMMING, HIGH LEVEL PROGRAMMING, COMPUTER ARCHITECTURE
Electrotechnics	N6	1 DC MACHINES 2 AC THEORY 3 TRANSFORMERS 4 AC MACHINES
Industrial Electronics	N6	1. X-Rays 2. Radio activity 3. electronic safety devices 4. Ultrasonics 5. Automatic inspection
Mathematics	N6	Chapter 1: Differentiation Chapter 2 : Integration Chapter 3 : Partial Fractions Chapter 4 :Differential Equations

ENGINEERING STUDIES

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Power Machines	N6	steam nozzles, steam planat, air compressor, internal combustion engines
Building Administration	N6	Module 5: Workstudy and performance Module 6: Occupational health and safety Module 7: Quality and ethics in construction Modukle 8: Contract law and labour law
Building and Structural Construction	N6	1. Concrete design : Beams (Rectangular , Flanged beam), Slab , stairs , columns and Foundation 2. Steel Design : Frames , Moment of Inertia ; Steel colums
Chemical Plant Operations	N6	*Level and Quality Measurement*.
Chemical technology	N6	*MODULE 1:drying and equipment, Module 2: crystallisation, MODULE 3: evaporation calculation(BPR) and theory, MODULE 4: 3 methods of heat transfer*
Engineering Physics	N6	*Module 1: SOUND, Module 2: THERMODYNAMICS,Module 3: ELECTROSTATICS , Module 4:ATOMIC PHYSICS*
Mechanical D& Design	N6	belt drives, gear drives, line shaft
Mechanotechnics	N6	Line shaft, flywheels, reduction gear box, brakes
Quantity Surveying	N6	Module 3.4: R.C. Staircase. Module 3.5: Concrete Piling. Module 4: Quantities & Costing Module 1 : Construction Processes Theory & Standard System Work
Instruments	N6	Intrinsic Safety,Automatic Control and valves,Steam Boilers
SUBJECT	**LEVEL**	**MODULE**
Building Administration	102	Module 5: Building materials Module 6: Support to structures Module 7: Scaffoldings Module 8: Lifts and cranes Module 9: Programming of work
Building and Structural Construction	102	1. Foundation and Basements 2. Brickwork 3. Archs 4. Steel work 5. Some Theory
Chemistry	102	Module 1: INTRODUCTION TO ORGANIC CHEMISTRY(Isomerism and nomenclature) , Module 2: HYDROCARBONS(PROPERTIES, PREPARATIONS AND REACTIONS : ALKANES,ALKENES,ALKYNES,ALCOHOLS, ALDEHYDES AND KETONES). MUST KNOW COMMON NAMES
Digits	102	FURTHER CODES MODULE 4: LOGIC GATES MODULE 5: BOOLEAN ALGEBRA MODULE 9: ELEMENTARY FLIP-FLOPS MODULE 10: COUNTERS
Electrotechnics	102	MODULE 2: DC Machines. DC Motors, DC Generators. MODULE 3: AC Circuits. RLC Circuits. MODULE 4: Transformers

ENGINEERING STUDIES

TEST 2 SCOPE 2026

Engineering Science	102	MODULE 1: KINEMATICS MODULE 2: ANGULAR MOTION MODULE 3: DYNAMICS MODULE 4 : STATICS MODULE 5: HYDRAULICS(HYDRAULIC PRESSES ONLY)
Industrial Electronics	102	SUPERPOSITION,AC CIRCUITS,POWER SUPPLIES,ELECTRONIC POWER CONTROL,OP AMPS
Mathematics	102	MODULE 4;TRIGONOMETRY MODULE 5;GRAPHS MODULE 6;LIMITS AND DIFFERENTIATION
Mechanical Draughting	102	Loci Cam Profiles Sectional Drawing Detailed Drawing
Mechanotechnics	102	Module 1: Workshop layout, Module 9 : Metal cutting machines Module 5 : Gear drives , Module 8 : Hydraulic systems
Quantity Surveying	102	Module 4 : SI Units & Abbreviations from projects, Module 4.2 : Superstructure Project Measurements or 4.6 Plain Opening Project Measurements, Module 5 : Working Up Process & Theory, Module 6 : Price Analysis & Cost Estimates.
Analytical Chemistry	102	Applications og gravimetric methods, Titrimetric methods of analysis, Volumetric calculations, gravimetric titrimetry.
Organic Chemistry	102	Chemical reactivity, Chemical reactions of alkanes and cycloalkanes, Isomerism - stereochemistry

ENGINEERING STUDIES
TEST 2 SCOPE 2026